Software Requirements Specification (SRS)

1. Purpose

Turn the approved PRD into testable software behavior. The existing repository's frozen schemas, SDK and tested contracts take precedence; reconcile this SRS with them before changing code.

2. Functional specification

Quote input

- Validate required fields before any portal session starts.
- Preserve entered values after validation errors.
- Identify insurer-specific missing fields.
- Use a stable quote ID and timestamp.
- If census import is not approved for v1, do not imply bulk Excel upload is included.

Session and orchestration

- One isolated session per insurer.
- Explicit connection states: not connected, connected, login required, OTP required, CAPTCHA/manual action required, blocked, running, completed, failed, cancelled, unsupported.
- Bounded concurrency and per-insurer timeout.
- At most one retry, only for classified transient errors and safe steps.
- Cancellation is idempotent.
- Partial success is retained.
- A portal security control must never be bypassed.

Results

- Normalize currency and premium into one canonical representation.
- Store result provenance: automated or manual.
- Never fabricate or default a missing premium to zero.
- Mark ambiguous parse results for review.
- Cheapest highlighting only when results meet comparable coverage assumptions.
- Advisor can review/exclude results before PDF if that feature is approved in v1/P1.

PDF

- Render from validated stored result data, not unvalidated UI values.
- Generate one-to-eight insurer layouts.
- Clearly distinguish automated, manual, failed, timed-out and unsupported results.
- Include approved disclaimer and branding.
- Report file-generation failure without losing the quote.

History and settings

- Local persistence and migrations.
- Search/reopen/duplicate/regenerate/delete.

- Deleting a quote also handles associated screenshots and app-managed PDFs according to policy.
- Retention period must be approved and documented.
- Secrets stored separately from normal application data.

3. Non-functional requirements

- Security: renderer isolation, minimal preload API, main-process validation, restricted navigation/popups, secure credential handling.
- Reliability: adapter failure isolation, safe cancellation, no false-success states, transactional related writes.
- Performance: test PRD targets under recorded conditions.
- Usability: errors explain next action; state not communicated by color alone.
- Compatibility: verify all target OS versions/architectures against actual runtime/native dependencies.
- Maintainability: adapter-specific parser tests and sanitized fixtures.
- Accessibility: keyboard navigation, focus management, readable contrast, selectable PDF text.
- Privacy: data minimization, safe logs, retention, deletion and export warnings.

4. External interfaces

- Approved insurer portals.
- Local filesystem for PDFs/evidence.
- SQLite or existing approved local store.
- OS-backed secret storage.
- Installer/signing/notarization/update services.

5. Requirement language

MUST = release requirement; SHOULD = recommended; MAY = optional. Any change to a P0 requirement requires product owner approval and traceability update.